

1. *How many entries do you have in your database that have applied for Fall 2026?*

I used COUNT(*) on the applicants table filtered to term = 'Fall 2026' to count how many rows match that term. This gives the total number of Fall 2026 applications in the database, which totaled 69,990.

2. *What percentage of entries are from international students (not American or Other) (to two decimal places)?*

I computed the share of rows where "us_or_international" is not in ('American', 'Other'), divided by the total count of rows with a non-null "us_or_international", then multiplied by 100 and rounded to 2 decimals. This estimates the percentage of international applicants using the category field, which gave me 50.12%.

3. *What is the average GPA, GRE, GRE V, and GRE AW of applicants who provide these metrics?*

I used AVG(...) on 'gpa', 'gre', 'gre_v', and 'gre_aw', but filtered to rows where all four metrics are present and > 0, so the averages aren't dragged by missing/zero placeholders. And I got a GPA of 3.79, GRE of 197.35, GRE V 159.82, and GRE AW 8.01.

4. *What is the average GPA of American students in Fall 2026?*

I filtered the table to "term = 'Fall 2026'" and average GPA (only where gpa>0). This isolates the GPA distribution for American applicants in that term, and came out to 4.07 GPA.

5. *What percent of entries for Fall 2026 are Acceptances (to two decimal places)?*

I computed the acceptance rate for Fall 2026 by counting rows with "status='Accepted'" and dividing by total Fall 2026 rows, then multiplying by 100 and rounding. This gives 24.32%.

6. *What is the average GPA of applicants who applied for Fall 2026 and were accepted?*

I averaged GPA for Fall 2026 entries where "status='Accepted'" and 'gpa>0'. This tells me the typical GPA among accepted applicants for that term, which was 3.8.

7. *How many entries are from applicants who applied to JHU for a masters degrees in Computer Science?*

I counted entries where the downloaded program text contains “Johns Hopkins” and “computer science”, and applied for a master’s degree. I was able to find only 6 entries.

8. *How many entries from 2026 are acceptances from applicants who applied to Georgetown University, MIT, Stanford University, or Carnegie Mellon University for a PhD in Computer Science?*

I counted Fall 2026 PhD applications toward programs from Georgetown, MIT, CMU, Stanford, in computer science, through “ILIKE”. I only got 2 entries that resulted in an offer.

9. *Do your numbers for question 8 change if you use LLM-generated fields (rather than your downloaded fields)?*

I repeated the steps of Q8, but instead of searching the raw downloaded program field for the university name, I used the LLM-generated fields. This checks whether the LLM normalization changes the count. Nothing changed.

10. *What are the top 10 programs with the highest acceptance rate?*

Grouping the Fall 2026 entries by program, I computed the total applications and accepted counts, then calculated accepted/total times 100 as a rate, filtering to programs with at least 25 entries using `HAVING COUNT(*) >= 25`. This was to avoid getting too much noise in my sample, from outliers, and to be careful not to set it too high that it is unrealistic. Top results were in Physics (U of Wisconsin-Madison) at a 15.79% rate, and so on.

11. *What is the average GRE of each status category?*

I grouped Fall 2026 entries by status and computed the average GRE Quant for rows with ‘gre>0’, then sorted them from highest to lowest (DESC). This compares the typical GRE Quant across outcome categories. The top score outcomes were: Waitlisted at 299.7, Interviewed at 276.14, Rejected at 272.42, and Accepted at 266.7

Console Output:

```

PS C:\Users\andre\code\jhu_software_concepts> cd module_3
PS C:\Users\andre\code\jhu_software_concepts\module_3> python query_data.py
Postgres password for user 'postgres':

=====
Q1) How many entries are in my database for Fall 2026?
-----
(6990,)

=====
Q2) What percent of entries are international (not American or Other)?
-----
(Decimal('50.12'),)

=====
Q3) Average GPA / GRE / GRE V / GRE AW (only rows that provide all metrics).
-----
(Decimal('3.79'), Decimal('197.35'), Decimal('159.82'), Decimal('8.01'))

=====
Q4) Average GPA of American students in Fall 2026.
-----
(Decimal('4.07'),)

=====
Q5) What percent of Fall 2026 entries are Acceptances?
-----
(Decimal('24.32'),)

=====
Q6) Average GPA of Fall 2026 Acceptances.
-----
(Decimal('3.80'),)

=====
Q7) How many entries applied to JHU for a masters in Computer Science? (downloaded fields)
-----
(6,)

=====
Q8) Fall 2026 acceptances for (Georgetown/MIT/Stanford/CMU) PhD CS (downloaded fields).
-----
(2,)

=====
Q9) Same as Q8 but using LLM-generated fields.
-----
(2,)

=====
Extra Q10) Top 10 programs by acceptance rate (Fall 2026, n>=200).
-----
('Physics, University of Wisconsin - Madison', 38, 6, Decimal('15.79'))
('Philosophy, University of Chicago', 28, 2, Decimal('7.14'))
('Chemistry, University of Wisconsin - Madison', 33, 2, Decimal('6.06'))
('Physics, University of Chicago', 37, 2, Decimal('5.41'))
('History, Yale University', 31, 1, Decimal('3.23'))
('Mathematics, University of Wisconsin - Madison', 39, 1, Decimal('2.56'))
('Political Science, University of Chicago', 32, 0, Decimal('0.00'))

=====
Extra Q11) Average GRE Quant by status (Fall 2026, gre>0).
-----
('Wait listed', Decimal('299.70'))
('Interview', Decimal('276.14'))
('Rejected', Decimal('272.42'))
('Accepted', Decimal('266.70'))

```

Andres' GradCafe Data Analysis

Update Analysis

Re-runs the SQL queries and refreshes the dashboard using the data already in PostgreSQL.

Pull Data Pull Data fetches the latest GradCafe results, processes them, and updates the analysis database.

Q1) How many entries do you have in your database that have applied for Fall 2026?

fall_2026_entries
6990

Q2) What percentage of entries are from international students?

pct_international
50.12

Q3) What is the average GPA, GRE, GRE V, and GRE AW of applicants who provide these metrics?

avg_gpa	avg_gre	avg_gre_v	avg_gre_aw
3.79	157.35	159.82	8.01

Q4) What is the average GPA of American students in Fall 2026?

round
4.07

Q5) What percent of entries for Fall 2026 are Acceptances?

round
24.32

Q6) What is the average GPA of applicants who applied for Fall 2026 and were accepted?

round
3.80

Q7) How many entries are from applicants who applied to JHU for a masters degrees in Computer Science?

count
6

Q8) How many entries from 2026 are acceptances from applicants who applied to Georgetown University, MIT, Stanford University, or Carnegie Mellon University for a PhD in Computer Science?

count
2

Q9) Do your numbers for question 8 change if you use LLM-generated fields (rather than your downloaded fields)?

count
2

```
PS C:\Users\andre\code\jhu_software_concepts\module_3> & "C:\Program Files\PostgreSQL\18\bin\psql.exe" -U postgres -d module_3db
```

```
module_3db=# SELECT
module_3db=#   llm_generated_university AS institution,
module_3db=#   program,
module_3db=#   term,
module_3db=#   status
module_3db=# FROM applicants
module_3db=# WHERE
module_3db=#   llm_generated_university ILIKE '%johns hopkins%'
module_3db=# OR llm_generated_university ILIKE '%hopkins%'
module_3db=# OR llm_generated_university ILIKE '%harvard%'
module_3db=# OR llm_generated_university ILIKE '%yale%'
module_3db=# OR llm_generated_university ILIKE '%princeton%'
module_3db=# OR llm_generated_university ILIKE '%columbia%'
module_3db=# OR llm_generated_university ILIKE '%brown%'
module_3db=# OR llm_generated_university ILIKE '%dartmouth%'
module_3db=# OR llm_generated_university ILIKE '%penn%'      -- UPenn usually matches here
module_3db=# OR llm_generated_university ILIKE '%cornell%'
module_3db=# ORDER BY term DESC
module_3db=# LIMIT 50;
```

institution	program	term	status
Yale University	Chemistry		Interview
Brown University	Earth, Environmental and Planetary Sciences	Fall 5	Rejected
Columbia University	Political Science	Fall 5	Accepted
Johns Hopkins University	Cell and Molecular Medicine	Fall 25	Accepted
Columbia University	Masters in Public Health	Fall 2026	Accepted
Johns Hopkins University	Sociology	Fall 2026	Interview
Johns Hopkins University	Chemical And Biomolecular Engineering	Fall 2026	Interview
university of pennsylvania	Ancient History	Fall 2026	Rejected
Pennsylvania State University	Materials Science And Engineering	Fall 2026	Interview
Brown University	Health Services Research	Fall 2026	Interview
Columbia University	Computer Science	Fall 2026	Accepted
Penn State University	Materials Science And Engineering	Fall 2026	Interview
Johns Hopkins Bloomberg School of Public Health	International Health (Human Nutrition)	Fall 2026	Interview
Johns Hopkins University	Sociology	Fall 2026	Interview
Cornell University	Mechanical And Aerospace Engineering	Fall 2026	Interview
Weill Cornell Medicine	Tri-Institutional Program in Chemical Biology	Fall 2026	Rejected